

Scalability & Future Plan

Date	2026-07-11
Team ID	XXXXXX
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

This document explains how the EduGenie – Google Gemini Powered Learning Assistant can be expanded to support a larger number of users, improve performance, enhance security, and introduce additional AI-powered learning features in future releases.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Supports a limited number of concurrent users.	Performance may decrease when many users access the application simultaneously.	High
2	No user login or progress tracking.	Users cannot save learning history or personalized progress.	High
3	Depends on the Google Gemini API and internet connection.	AI features are unavailable if the API or network is unavailable.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local FastAPI application with limited users.	Deploy on cloud platforms with multi-user support and load balancing.
2	Data Storage	No permanent database is used.	Integrate MongoDB to store user profiles, quizzes, summaries, and learning progress.
3	Performance	AI requests are processed through the Google Gemini API.	Optimize API requests, asynchronous processing, and caching.
4	Security	Basic API key integration.	Add authentication, HTTPS, encrypted API keys, and role-based access control.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	User Login & Progress Tracking	2 Months	Personalized learning experience with saved progress.
Phase 2	MongoDB Database Integration	4 Months	Reliable and scalable data storage.

Phase 3	Voice-Based Learning & Multilingual Support	6 Months	Improved accessibility and wider user adoption.
Phase 4	Mobile Application & Cloud Deployment	8 Months	Better scalability and accessibility.